Orbits

Notes on calculations on classical orbits under gravity

Hannah Michelle Ellis

November 19, 2025

Contents

List of Tables

List of Figures

1 Plane Polar Coordinate System

In this section we will derive the equations of motion for an object acted upon by gravity in plane polar coordinates. To do this we will first consider plane polar coordinates themselves and the derivatives of the position vector within the coordinate system.

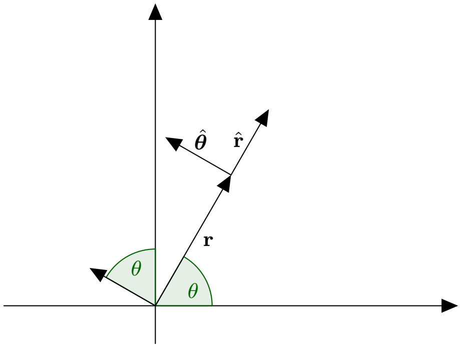

Figure 1: Position and unit vectors in plane polar coordinates

1.1 Polar unit vectors

The unit vectors of plane polar coordinates can be determined from the standard unit vectors as

$$\hat{\boldsymbol{r}} = \hat{\mathbf{i}}\cos(\theta) + \hat{\mathbf{j}}\sin(\theta) \tag{1.1}$$

$$\hat{\boldsymbol{\theta}} = -\hat{\mathbf{i}}\sin(\theta) + \hat{\mathbf{j}}\cos(\theta) \tag{1.2}$$

Where r is the distance from the origin to the position or alternatively the length of the position vector, and θ is the angle between $\hat{\mathbf{i}}$ and $\boldsymbol{r}$. Since the unit vectors $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{\theta}}$ depend on θ which changes with time, the unit vectors also change with time. We will begin by looking at the derivative of $\hat{\boldsymbol{r}}$ with respect to time.

$$\frac{d\hat{\boldsymbol{r}}}{dt} = \frac{d\theta}{dt}\frac{d}{d\theta}\hat{\boldsymbol{r}} = \frac{d\theta}{dt}\left[-\hat{\mathbf{i}}\sin(\theta) + \hat{\mathbf{j}}\cos(\theta)\right] = \frac{d\theta}{dt}\hat{\boldsymbol{\theta}} \tag{1.3}$$

where we have used equation 1.2 to replace the typical Cartesian coordinate unit vectors.

$$\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d\theta}{dt}\frac{d}{dt}\hat{\boldsymbol{\theta}} = \frac{d\theta}{dt}\left[-\hat{\mathbf{i}}\cos(\theta) - \hat{\mathbf{j}}\sin(\theta)\right] = -\frac{d\theta}{dt}\hat{\boldsymbol{r}} \tag{1.4}$$

Since $\frac{d\hat{\boldsymbol{r}}}{dt}$ too depends on θ so is also time dependent.

$$\frac{d^2\hat{\boldsymbol{r}}}{dt^2} = \frac{d}{dt}\left[\frac{d\theta}{dt}\hat{\boldsymbol{\theta}}\right] = \frac{d^2\theta}{dt^2}\hat{\boldsymbol{\theta}} + \frac{d\theta}{dt}\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d^2\theta}{dt^2}\hat{\boldsymbol{\theta}} - \left(\frac{d\theta}{dt}\right)^2\hat{\boldsymbol{r}} \tag{1.5}$$

Polar unit vectors summary

The polar coordinate unit vectors are

$$\hat{\boldsymbol{r}} = \hat{\mathbf{i}}\cos(\theta) + \hat{\mathbf{j}}\sin(\theta)$$
$$\hat{\boldsymbol{\theta}} = -\hat{\mathbf{i}}\sin(\theta) + \hat{\mathbf{j}}\cos(\theta)$$

With the first derivatives being

$$\frac{d\hat{\boldsymbol{r}}}{dt} = \omega\hat{\boldsymbol{\theta}}$$
$$\frac{d\hat{\boldsymbol{\theta}}}{dt} = -\omega\hat{\boldsymbol{r}}$$

And the second derivative of $\hat{\boldsymbol{r}}$ being

$$\frac{d^2\hat{\boldsymbol{r}}}{dt^2} = \frac{d\omega}{dt}\hat{\boldsymbol{\theta}} - \omega^2\hat{\boldsymbol{r}}$$

with $\omega = \frac{d\theta}{dt}$

1.2 Velocity and acceleration in polar coordinates

Here we will calculate the acceleration in terms of the polar unit vectors.

1.2.1 Position

Starting with the position vector

$$\mathbf{r} = r\hat{\mathbf{r}} \tag{1.6}$$

This is the length of the position vector in the direction of $\hat{\boldsymbol{r}}$.

1.2.2 Velocity

The first derivative will be the velocity of the position vector in terms of the polar unit vectors

$$\frac{d\boldsymbol{r}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\frac{d\hat{\boldsymbol{r}}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}} \tag{1.7}$$

The velocity is the sum of the radial velocity in the radial direction $\dot{r}$ and the angular part of the velocity $r\omega$ in the angular direction.

1.2.3 Acceleration

$$\begin{aligned}
\frac{d^2\boldsymbol{r}}{dt^2} &= \frac{d^2r}{dt^2}\hat{\boldsymbol{r}} + \frac{dr}{dt}\frac{d\hat{\boldsymbol{r}}}{dt} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + r\frac{d\omega}{dt}\hat{\boldsymbol{\theta}} + r\omega\frac{d\hat{\boldsymbol{\theta}}}{dt} \\
&= \frac{d^2r}{dt^2}\hat{\boldsymbol{r}} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + r\frac{d\omega}{dt}\hat{\boldsymbol{\theta}} - r\omega^2\hat{\boldsymbol{r}} \\
&= \left[\frac{d^2r}{dt^2} - r\omega^2\right]\hat{\boldsymbol{r}} + \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt}\right]\hat{\boldsymbol{\theta}}
\end{aligned} \tag{1.8}$$

1.2.4 Acceleration in polar coordinates summary

The position vector

$$\boldsymbol{r} = r\hat{\boldsymbol{r}}$$

Velocity

$$\frac{d\boldsymbol{r}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}$$

Acceleration

$$\frac{d^2\boldsymbol{r}}{dt^2} = \left[\frac{d^2r}{dt^2} - r\omega^2\right]\hat{\boldsymbol{r}} + \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt}\right]\hat{\boldsymbol{\theta}}$$